CAPADEV ENGINEERING CHALLENGE · PROJECT BRIEF

PulseTrack

A remote monitoring platform for a small diabetes clinic — so that the eight to twelve weeks between appointments stop being a blind spot.

USERS	DATA	INTEGRATES WITH	DELIVERY
Clinicians only. Patients never get an account.	Lab results, self-assessment scores, patient records.	A national health data platform over FHIR R4.	Deployed live on Vercel, submitted by Wed 26 Aug.

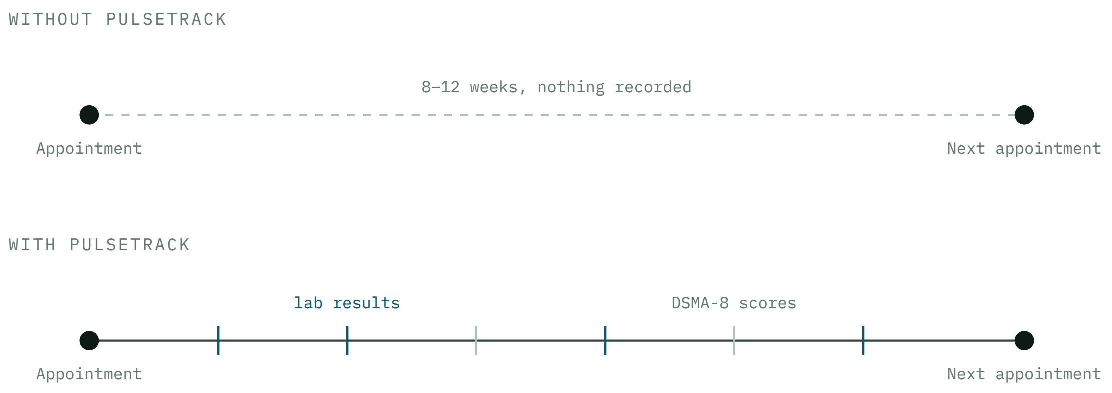
THE PROBLEM

A clinic can only see a patient on the days it sees them

Type 2 diabetes is managed almost entirely by the patient, at home, every day. The clinic's contribution arrives in short bursts — an appointment every couple of months. Everything that actually determines the outcome happens in between.

Two kinds of information are generated during that gap, and neither of them reaches the clinician until the next visit. **Lab results** are drawn at a lab and come back as a file. **Self-management** — whether doses were taken, glucose was checked, feet were inspected, whether the patient is coping at all — is only ever recalled from memory at the appointment, which is exactly when it is least reliable.

PulseTrack's whole purpose is to fill that window with real, dated, structured data, so that a trend already exists on screen when the patient walks in.



The same interval, before and after. **Nothing about the appointments changes** — what changes is that the clinician arrives at the second one already holding a dated series instead of a blank chart and a patient's recollection.

THE PRODUCT

Four things the clinic can do that it could not before

Everything in Tier 1 exists to serve one of these four loops. If a feature does not feed one of them, it is not in scope.

01

Keep a patient register

Name, date of birth, sex, a unique medical record number, and contact details. The MRN is the key everything else hangs off — labs, assessments, and the national platform all identify a person by it.

02

Ask the patient how it is going

Email a one-off link to an eight-question self-assessment. The patient answers without ever creating an account. The system scores it and places them in a risk band.

03

Load lab results in bulk

Results arrive from a lab as a file, not through the app. Upload the CSV, get a row-by-row verdict, and keep every valid row even when other rows are wrong.

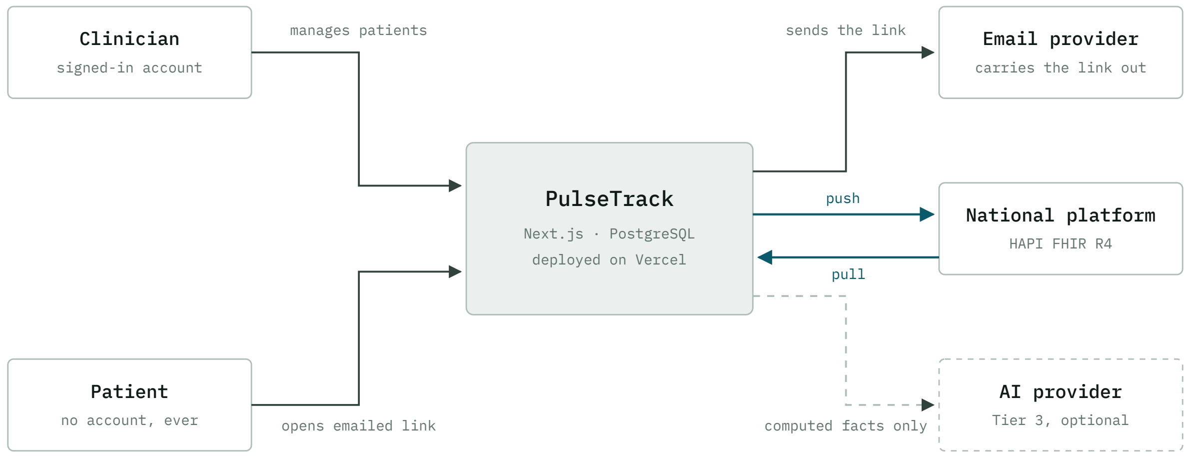
04

See the trend, not the number

Per patient: glucose, HbA1c and assessment scores over time. Across the clinic: how many patients, how many are answering, and who has drifted into a higher risk band.

Who touches the system

The unusual constraint here is the second actor. A patient must be able to answer a questionnaire, but the brief forbids patient accounts anywhere in the system. That single rule shapes the entire assessment design.



Every arrow leaving the box is a place data can leak or a call can fail. The email provider carries a live credential out of the system; the national platform is shared with other candidates and enforces ownership; the AI provider only ever receives numbers we computed ourselves, never a patient's identity.

DOMAIN

The three measurements, and why it is these three

The challenge data — both the CSV template and the twelve months of seeded history on the FHIR server — contains exactly three tests. That is not arbitrary. Between them they cover long-term control, short-term control, and the thing that actually causes harm.

TEST	CODE	REFERENCE	WHAT IT TELLS THE CLINICIAN
Hemoglobin A1c LOINC 4548-4	HBA1C	4.0–5.6 %	The share of haemoglobin that has glucose stuck to it. Red cells live about three months, so this is an *average* of the last 8–12 weeks. It is the anchor measure, and it cannot be improved by behaving well for the week before the appointment.
Fasting glucose LOINC 1558-6	GLU-F	70–99 mg/dL	A single reading after eight hours without food. Volatile from day to day — genuinely informative as a series, close to meaningless as one number, which is why the dashboard plots it rather than displaying it.
Systolic blood pressure LOINC 8480-6	SBP	90–120 mmHg	Diabetes does its damage through blood vessels — heart, kidneys, eyes, nerves. Blood pressure is tracked next to glucose because it is the second lever on the same complications.

WHY THIS MATTERS FOR THE BUILD

These three codes are the entire test vocabulary. A CSV row carrying anything else is rejected, and each maps to exactly one LOINC code when pushed to the national platform. Keeping the vocabulary closed is what makes both the import validation and the FHIR mapping tractable.

CAPABILITY 02

DSMA-8, and what it is actually measuring

The Diabetes Self-Management Assessment is a fictional instrument written for this challenge, but it is modelled on real ones and it is internally coherent. Eight statements, each answered on a four-point frequency scale from *Never* (0) to *Nearly every day* (3), covering the last two weeks.

Every item is negatively framed — each one describes something going wrong — so a **higher score is worse** and there is no reverse scoring to get wrong. Reading the eight items together, they sample four distinct failure modes:

ROUTINE ADHERENCE

1. I missed or skipped a dose of my diabetes medication.
2. I did not check my blood glucose when I was supposed to.
7. I did not check my feet for cuts, blisters, or sores.

Is the daily protocol actually happening? Foot checks sit here because diabetic neuropathy means a wound can go unnoticed until it is serious.

LIFESTYLE

3. I ate meals or snacks that I know are bad for my glucose control.
6. I went the whole day without any physical activity or walking.

The two behaviours that move glucose most directly, and the two hardest to sustain.

SYMPTOMS

4. I felt unusually thirsty or needed to urinate more often than normal.
5. I felt shaky, sweaty, or dizzy in a way that felt like low blood sugar.

Item 4 describes glucose running high; item 5 describes it running low. Together they catch instability in either direction — which the lab results, taken weeks apart, would miss entirely.

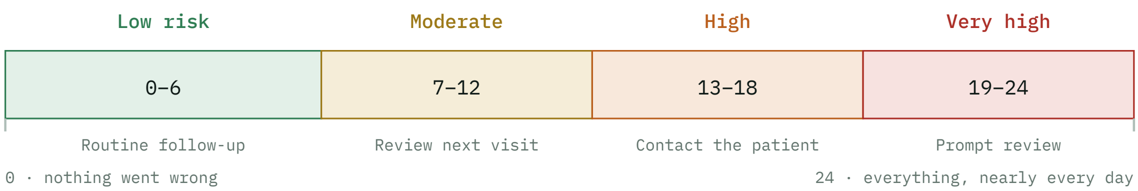
BURDEN

8. I felt too overwhelmed or discouraged to manage my diabetes.

The one item that predicts the other seven. A patient who has stopped coping will fail every category, and this is the only signal that arrives before the lab results do.

Scoring

Sum the eight answers. All eight are required, so the total is always between 0 and 24, and it falls into one of four bands that carry a recommended clinical response.



The bands are contiguous and cover the full range — **every possible total has exactly one band**, with the boundaries at 6/7, 12/13 and 18/19. Those six numbers are the first thing an evaluator will test.

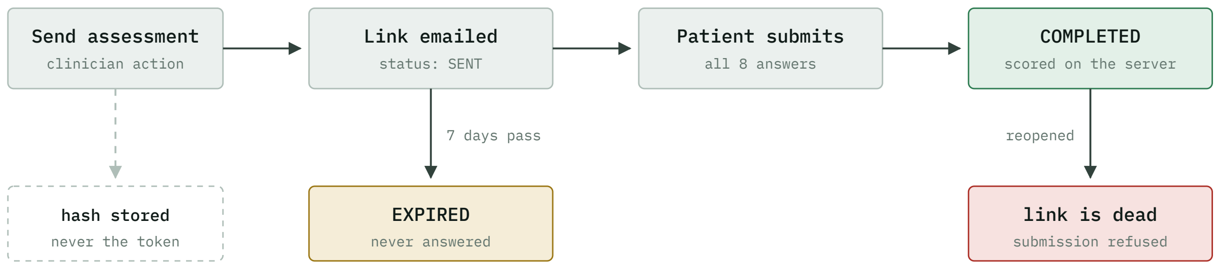
SECURITY BY DESIGN

The link is the login

Because patients have no account, there is nothing to authenticate them *with*. Whoever holds the link can answer the questionnaire — the URL itself is the credential. That makes three properties non-negotiable rather than nice to have.

- **Unguessable.** The token is 32 bytes of cryptographic randomness. Anything derived from a patient id, an MRN, or a counter would let someone walk the range and read other people's forms.
- **Short-lived.** It stops working seven days after it is sent, whether or not anyone used it.
- **Single-use.** Once submitted, the link is dead. A completed assessment is a clinical record; it must not be silently overwritten by whoever opens the email next.

One more property is ours rather than the brief's: we store only a SHA-256 hash of the token. The clinic's database therefore never contains a working link, so a database dump does not hand out access to outstanding questionnaires.



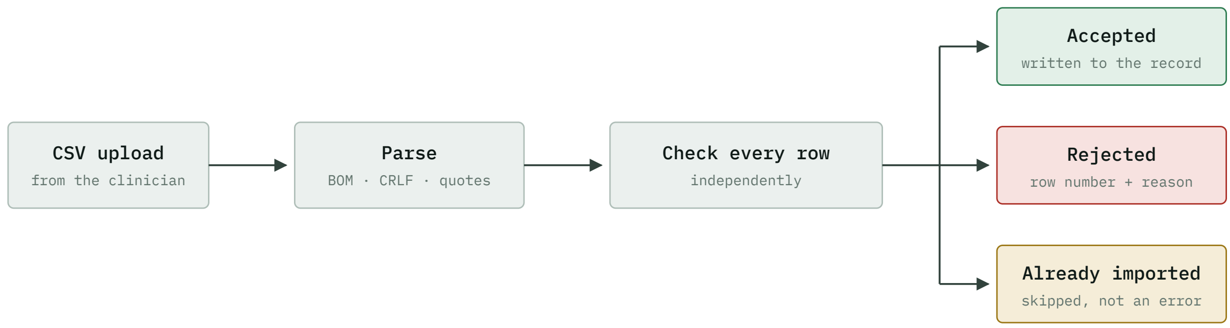
Both bottom-row states are terminal. The score is computed on the server from the stored answers — a total submitted by the browser is ignored entirely, because in a clinical record the number has to be one we derived, not one we were handed.

CAPABILITY 03

Lab results arrive as a file, and the file is usually a mess

Labs do not use the clinic's app. Results come back as a spreadsheet, and by the time one reaches the clinic it has been opened in Excel, re-saved, partially corrected, and emailed twice. The brief is explicit about this: *"We will test your uploader with a deliberately messy file. Ugly data is the point."*

So the import is not really a parsing feature. It is a feature about **how you behave when the input is wrong**, and the design rests on one rule: a bad row must never cost the clinic a good one. A two-hundred-row file with three broken rows imports one hundred and ninety-seven results and explains the other three.



Three outcomes, not two. Separating *already imported* from *rejected* is what makes re-uploading a corrected file read correctly: the rows that were fine come back as skipped, the rows that were broken now import, and nothing is duplicated.

What counts as a broken row

An MRN the clinic has never heard of. A date that is malformed, impossible, or in the future. A value that is not a number. A test code outside the three we recognise. A missing required field. And a row identical to one already on file — same patient, same date, same test — which is the definition of a duplicate the brief gives us.

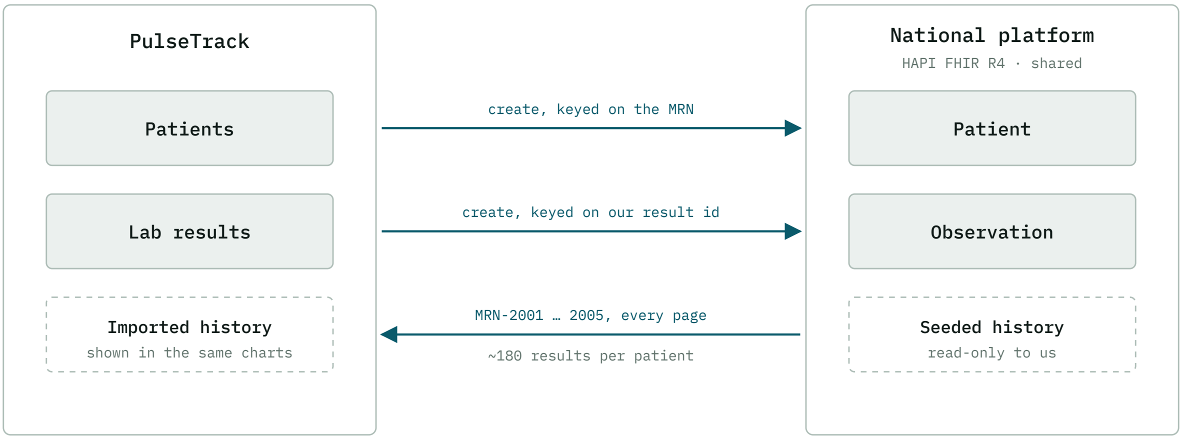
> THE RULE UNDERNEATH
>
> Duplicate detection is enforced in the database, not just in the import code, by a uniqueness constraint on patient + date + test. Application checks catch the common case and produce a readable message; the constraint is what makes the guarantee true even if two uploads run at the same moment.

TIER 2

A clinic does not hold its patients' whole history

Real patients move between providers. Their labs are run somewhere else, their previous clinic holds twelve months of readings, and the health service wants the clinic's data reported centrally. **FHIR** is the standard that makes that exchange possible — a specification for representing healthcare concepts as JSON resources, of which we need exactly two: `Patient` and `Observation`.

Tier 2 makes the clinic a participant rather than an island. It sends what it creates and receives what already exists, and the seeded history that comes back has to appear in the same charts as locally-entered results — otherwise it is an integration demo rather than a working platform.



Every write is conditional on a key we control — the MRN for a patient, our own result id for an observation. That is what makes the whole exchange safe to run twice: a repeated push finds what the first one created instead of creating a second copy, even when the first attempt's response was lost in transit.

The complication worth knowing about

The server is shared with every other candidate, and it enforces ownership: anyone can read anything, but you may only modify what you created. Seed data and other people's records are read-only. That is realistic — a national platform is exactly like this — and it means the app has to distinguish records it owns from records it merely holds a copy of, and refuse to attempt writes it knows will be refused.

SCOPE

Three tiers, in the order they earn their place

The brief is explicit that a complete, polished Tier 1 is a fully valid submission, and that quality beats quantity. The tiers are therefore a priority order, not a checklist — each one only starts when the one before it is finished and deployed.

Tier 1 required	**The clinic can run on it** Login, patient records, the assessment loop, bulk lab import, and both dashboards — with the loading, empty and error states treated as part of the work rather than an afterthought. This alone is a complete submission.
Tier 2 the differentiator	**The clinic is connected to the wider system** Two-way exchange with the national platform, handled like a real external dependency: authentication, rate limits, failures, retries, and — the part that is actually being assessed — never creating the same record twice.
Tier 3 only if safe	**One AI feature, judged on restraint** The brief asks for judgment, not ambition. The plan is a short trajectory summary where every number is computed in code and the model only writes the prose — so a fabricated figure is trivially detectable, and the clinician can check the summary against the facts printed beside it.

THE TRADE BEING MADE

The interview is Thursday at noon, which leaves roughly two and a half working days. Tier 3 is the only item that can be dropped without failing a stated requirement, so it is the buffer. The decision to attempt it gets made on Wednesday afternoon, once Tier 2 is deployed and the README is written — not before.

BOUNDARIES

What PulseTrack deliberately is not

Naming these is part of the design. Several would be obvious additions to a monitoring platform, and each one was left out for a reason worth being able to state.

- Patient accounts — forbidden by the brief, and the reason the tokenized link exists at all
- Appointment scheduling
- Prescribing or medication management
- Billing and insurance
- Clinician–patient messaging
- Live device or CGM feeds
- Multiple clinics or tenancy
- Clinical decision support beyond the questionnaire's own guidance
- An audit log — genuinely required in production, out of scope at this deadline
- Any claim of HIPAA or GDPR compliance

The last two are the honest ones. Good security practice is not the same as regulatory compliance, and saying so is more credible than implying otherwise.

THE WALKTHROUGH

Questions this design should be ready to answer

The call is a demo followed by detailed questions about the code and the decisions. These are the ones the architecture invites.

Why is there no patient login?

Because the brief forbids it — and because it is the better design anyway. An account is a standing credential a patient would forget and reuse; a single-use link that expires in seven days grants exactly the access needed, once, and then stops existing.

Why store a hash of the token rather than the token?

The token is a live credential. Storing it means the database contains working links to outstanding questionnaires. Storing only the hash means a leaked dump grants nothing, and costs one hash per lookup.

Why compute the score on the server when the browser already knows it?

Because the browser is untrusted input and the score becomes a clinical record. The submitted total is ignored; the score is derived from the stored answers by the same pure function the tests exercise at every band boundary.

Why does one bad CSV row not fail the file?

Because rejecting 200 good results over 3 bad ones makes the feature useless in exactly the situation it exists for. Rows are validated independently, and the report explains each rejection by row number.

What stops a re-uploaded file from duplicating results?

A uniqueness constraint on patient + date + test in the database. The import code checks first so the message is readable, but the constraint is what makes the guarantee hold under concurrency.

How is the FHIR push safe to retry?

Every create is conditional on a key we control, so a retry matches what the first attempt created instead of creating a second copy. This matters most in the ambiguous case — where the server succeeded but the response never came back.

What happens when the national platform is down?

Nothing that was already saved is lost. Local writes commit first and sync is a separate, recorded step, so a failed push leaves a lab result in the record with a visible sync state rather than rolling it back.

Why does the AI feature only narrate precomputed numbers?

Because the failure mode in a clinical context is a plausible invented figure. If every number in the prose already exists in the input, a fabrication is detectable — and the facts are printed next to the summary so the clinician can check it in a glance.

What would you do differently with more time?

An audit log, an outbox for the integration rather than inline sync, structured patient names instead of a single field, and a background job for assessment expiry rather than deriving it on read. Each is a deliberate trade, not an oversight.